

Towards a mathematical theory of science

1 Overview

We propose a formal framework for representing scientific discourse as trajectories over a semantic embedding space. The framework distinguishes carefully between:

- (i) empirical objects constructed directly from data,
- (ii) statistical structures inferred from empirical regularities,
- (iii) and theoretical interpretations imposed on top of those structures.

The framework does not assume that embeddings directly represent the true geometry of knowledge. Instead, embeddings are treated as noisy coordinate systems induced by statistical regularities in scientific language.

2 Representation Layer

2.1 Embedding Model

Let

$$E : \mathcal{T} \rightarrow \mathbb{R}^d$$

be a semantic embedding model mapping text segments into a d -dimensional vector space. Here:

- $\mathcal{T}$ denotes the space of textual expressions,
- E is a pretrained embedding function,
- $\mathbb{R}^d$ is the embedding coordinate space.

Importantly, $\mathbb{R}^d$ is not assumed to be the ontology of scientific knowledge itself. Rather, it is an empirical representation space induced by the statistical structure of language.

2.2 Discourse Segmentation

For each scientific document i , we extract discourse-role segments:

$$\tau_i^{text} = (t_i^P, t_i^M, t_i^F, t_i^I)$$

where:

- t_i^P is the problem discourse segment,
- t_i^M is the method discourse segment,

- t_i^F is the finding discourse segment,
- t_i^I is the interpretation discourse segment.

These segments are rhetorical or discourse-functional partitions extracted from scientific texts.

2.3 Discourse-Role Embeddings

Applying the embedding model to each segment yields:

$$p_i = E(t_i^P)$$

$$m_i = E(t_i^M)$$

$$f_i = E(t_i^F)$$

$$i_i = E(t_i^I)$$

with:

$$p_i, m_i, f_i, i_i \in \mathbb{R}^d$$

The tuple:

$$\tau_i = (p_i, m_i, f_i, i_i)$$

defines the embedded discourse trajectory associated with document i .

3 Empirical State Distributions

We define empirical discourse distributions over discourse-role embeddings:

$$P = \{p_i\}_{i=1}^N$$

$$M = \{m_i\}_{i=1}^N$$

$$F = \{f_i\}_{i=1}^N$$

$$I = \{i_i\}_{i=1}^N$$

These are not assumed to be independent ontological spaces. Instead, they are empirical distributions induced by different discourse functions over the same embedding geometry.

Thus:

$$P, M, F, I \subset \mathbb{R}^d$$

represent semantically conditioned regions of the embedding space.

4 Observed Discourse Trajectories

A scientific document induces an ordered sequence:

$$\tau_i = (p_i, m_i, f_i, i_i)$$

which we interpret empirically as an observed discourse trajectory.

Importantly, this trajectory does not necessarily represent literal motion through an underlying epistemic manifold. Instead, it represents a sequence of semantically related discourse states co-occurring within a scientific reasoning process.

5 Local Displacement Operators

5.1 Definition

For each document i , we define local displacement vectors:

$$\Delta_{PM}^{(i)} = m_i - p_i$$

$$\Delta_{MF}^{(i)} = f_i - m_i$$

$$\Delta_{FI}^{(i)} = i_i - f_i$$

where:

$$\Delta_{PM}^{(i)}, \Delta_{MF}^{(i)}, \Delta_{FI}^{(i)} \in \mathbb{R}^d$$

These vectors are empirical displacement estimates between discourse-role embeddings.

5.2 Interpretation

The displacement vectors are not assumed to be literal epistemic operators. Instead, they are interpreted as:

semantic displacement estimates associated with transitions between discourse roles within scientific documents.

For example:

$$\Delta_{PM}^{(i)}$$

captures the semantic displacement associated with moving from problem discourse to method discourse within document i .

Similarly:

- Δ_{MF} captures transitions from methods to findings,
- Δ_{FI} captures transitions from findings to interpretation.

5.3 Operator Distributions

We define empirical displacement distributions:

$$\cdot_{PM} = \{\Delta_{PM}^{(i)}\}_{i=1}^N$$

$$\cdot_{MF} = \{\Delta_{MF}^{(i)}\}_{i=1}^N$$

$$\cdot_{FI} = \{\Delta_{FI}^{(i)}\}_{i=1}^N$$

These distributions characterize recurrent semantic transition patterns within scientific discourse.

6 Neighborhood Structure

Let:

$$N_k(x)$$

be the k -nearest-neighbor neighborhood of point x under a similarity metric $d(\cdot, \cdot)$. Typically:

$$d(x, y) = 1 - \cos(x, y)$$

or Euclidean distance after normalization.

Neighborhoods define local empirical association structure in embedding space.

7 Conditional Transition Distributions

7.1 Definition

We define conditional transition distributions between discourse-role states.

For example:

$$\mathcal{D}(M \mid p)$$

denotes the empirical distribution of method embeddings associated with problems near p . Operationally:

$$\mathcal{D}(M \mid p) = \{m_j \mid p_j \in N_k(p)\}$$

Similarly:

$$\mathcal{D}(P \mid m) = \{p_j \mid m_j \in N_k(m)\}$$

$$\mathcal{D}(F \mid m) = \{f_j \mid m_j \in N_k(m)\}$$

$$\mathcal{D}(I \mid f) = \{i_j \mid f_j \in N_k(f)\}$$

and so forth.

7.2 Interpretation

These distributions are not displacement vectors. They are conditional empirical distributions over neighboring discourse states.

Thus:

$$\mathcal{D}(Y \mid x)$$

should be interpreted as:

The empirical distribution of discourse states of type Y associated with semantic neighborhoods around state x .

8 Relationship Between Δ and $\mathcal{D}$

The framework distinguishes two fundamentally different empirical objects.

8.1 Observed Local Displacements

$$\Delta_{XY}^{(i)} \in \mathbb{R}^d$$

represents:

a single observed displacement between discourse-role embeddings within document i .

These are trajectory-local quantities.

8.2 Conditional Transition Structure

$$\mathcal{D}(Y \mid x)$$

represents:

The empirical distribution of possible neighboring discourse states associated with semantic region x .

These are population-level conditional structures.

9 Hypothesis Structure

The framework proposes three progressively stronger empirical hypotheses.

- H1: local discourse displacements exhibit non-random structure,
- H2: conditional transition distributions exhibit organized local geometry,
- H3: transition structure possesses predictive value for future semantic occupation patterns.

These hypotheses correspond respectively to:

- (i) existence of structured transition dynamics,
- (ii) organization of local transition landscapes,
- (iii) temporal predictive utility.

10 H1 — Structural Transition Hypothesis

10.1 Informal Statement

Scientific discourse transitions exhibit non-random geometric structure distinguishable from randomized baselines.

10.2 Formal Statement

Let:

$\cdot_{obs}$

be an observed displacement distribution and:

$\cdot_{null}$

be a displacement distribution generated under a null model.

H1 states that:

$\cdot_{obs} \not\sim \cdot_{null}$

under suitable geometric statistics.

10.3 Structural Metrics

Examples of structural statistics include:

- displacement norm distributions,
- cosine-angle distributions,
- local density distributions,
- anisotropy measures,
- neighborhood consistency,
- PCA variance spectra.

These quantities characterize geometric regularities of observed discourse transitions.

10.4 Null Models

Null distributions may be generated through:

- (a) random permutation of discourse segments,
- (b) shuffled document-role assignments,
- (c) randomized trajectory pairing,
- (d) synthetic corpora,
- (e) lexically perturbed corpora,
- (f) embedding-space randomization.

These models preserve varying degrees of lexical or embedding structure while destroying trajectory-level organization.

10.5 Statistical Test

For each metric distribution, we compare:

$$S_{obs}$$

against:

$$S_{null}$$

using the two-sample Kolmogorov–Smirnov statistic:

$$D_{KS} = \sup_x |F_{obs}(x) - F_{null}(x)|$$

where:

- F_{obs} is the empirical cumulative distribution function of observed statistics,
- F_{null} is the cumulative distribution under the null model.

The associated p-value tests:

$$H_0 : F_{obs} = F_{null}$$

against:

$$H_A : F_{obs} \neq F_{null}$$

Rejecting H_0 supports the existence of non-random transition structure.

11 H2 — Local Transition Organization Hypothesis

11.1 Informal Statement

Conditional transition distributions exhibit organized local geometry beyond random neighborhood structure.

11.2 Formal Statement

For a conditional transition distribution:

$$\mathcal{D}(Y \mid x)$$

H2 states that local transition organization differs significantly from randomized neighborhood structure.

11.3 Transition Geometry

For each semantic region x , we estimate:

$$\rho_x(y)$$

representing the empirical local transition density from states near x toward states of type Y .

These local distributions define a transition landscape over embedding space.

11.4 Null Models

Null transition distributions are constructed through:

- (a) random permutation of neighborhood assignments,
- (b) random rewiring of k-nearest-neighbor graphs,
- (c) shuffled discourse-role correspondences,
- (d) randomized conditional pairings.

These preserve global embedding geometry while destroying local transition organization.

11.5 Wasserstein Comparison

Let:

$$\rho_{obs}$$

be the observed local transition distribution and:

$$\rho_{null}$$

be the corresponding null distribution.

We compute the Wasserstein distance:

$$W(\rho_{obs}, \rho_{null})$$

which measures the minimum transport cost required to transform one distribution into the other.

Large Wasserstein distances indicate strong local organizational divergence from random structure.

11.6 Permutation-Based Significance

To estimate significance, we repeatedly generate randomized neighborhood structures:

$$\rho_{null}^{(1)}, \dots, \rho_{null}^{(B)}$$

and compute:

$$W^{(b)} = W(\rho_{obs}, \rho_{null}^{(b)})$$

The empirical p-value is:

$$p = \frac{1 + \sum_{b=1}^B \mathbf{1}[W^{(b)} \geq W_{obs}]}{B + 1}$$

where:

$$W_{obs}$$

is the Wasserstein distance obtained from the observed data.

Rejecting the null hypothesis supports the existence of organized local transition structure.

12 H3 — Predictive Transition Hypothesis

12.1 Informal Statement

Historical transition structure extracted from scientific discourse predicts future semantic occupation patterns better than baseline forecasting models.

12.2 Temporal Construction

Let:

$$P_t$$

represent the empirical problem-space distribution up to time t .
Similarly define:

$$M_t, F_t, I_t$$

and corresponding conditional transition structures:

$$\mathcal{D}_t(Y \mid x)$$

constructed exclusively from historical data available before time t .
Using these structures, we estimate a predicted future occupation density:

$$\hat{\rho}_{t+\Delta t}(p)$$

representing expected future semantic occupation in problem space.
Observed future occupation is defined as:

$$\rho_{t+\Delta t}(p)$$

constructed from papers appearing after time t .

12.3 Predictive Baselines

Predictions are compared against simpler baseline models.

Examples include:

(a) persistence models:

$$\hat{\rho}_{t+\Delta t}^{base} = \rho_t$$

(b) random diffusion models,

(c) nearest-neighbor continuation models,

(d) isotropic local expansion models.

These baselines capture generic smoothness or temporal persistence without discourse-transition structure.

12.4 Predictive Wasserstein Metric

We evaluate predictive accuracy through Wasserstein transport distance:

$$W(\hat{\rho}_{t+\Delta t}, \rho_{t+\Delta t})$$

Lower Wasserstein distance indicates stronger predictive alignment between predicted and observed future occupation.

H3 predicts:

$$W_{EDEL} < W_{baseline}$$

for appropriate baseline predictors.

12.5 Spatial Predictive Alignment

In addition to global transport similarity, we evaluate local spatial correspondence between predicted and observed density change.

Let:

$$\Delta\hat{\rho}_i$$

and:

$$\Delta\rho_i$$

represent predicted and observed density variation over semantic regions indexed by i .

We compute Bivariate Moran's I:

$$I_B = \frac{n}{W} \frac{\sum_i \sum_j w_{ij} (x_i - \bar{x})(y_j - \bar{y})}{\sum_i (x_i - \bar{x})^2}$$

where:

- $x_i = \Delta\hat{\rho}_i$,
- $y_j = \Delta\rho_j$,
- w_{ij} encodes semantic neighborhood adjacency,
- $W = \sum_{ij} w_{ij}$.

Positive Bivariate Moran's I indicates that regions predicted to expand correspond systematically to regions that actually expand.

12.6 Significance Estimation

Significance is estimated through temporal or spatial permutation procedures.

Examples include:

- (a) temporal shuffling of publication years,
- (b) randomized density assignments,
- (c) semantic-region permutation,
- (d) randomized baseline forecasts.

For each permutation:

$$I_B^{(b)}$$

is recomputed, generating an empirical null distribution.

The p-value is estimated as:

$$p = \frac{1 + \sum_{b=1}^B \mathbf{1}[I_B^{(b)} \geq I_{obs}]}{B + 1}$$

where:

$$I_{obs}$$

is the observed Bivariate Moran statistic.

Rejecting the null hypothesis supports predictive spatial alignment between transition-derived forecasts and future semantic occupation.

13 Novelty and Closure Metrics

13.1 Cycle Closure Approximation

A complete epistemic cycle may be interpreted theoretically as a transformation:

$$\Phi : P \rightarrow P$$

mapping an initial problem state into a future problem state generated through scientific reasoning.

Empirically, however, future problems are not directly observed within a single scientific abstract. Therefore, we define a first-order empirical approximation based on the semantic displacement between the initial problem embedding and the interpretation embedding.

For document i :

$$\Delta_{cycle}^{(i)} = i_i - p_i$$

Since:

$$(i_i - f_i) + (f_i - m_i) + (m_i - p_i) = i_i - p_i$$

this quantity represents the net open-cycle semantic displacement accumulated along the observed discourse trajectory.

13.2 Closure Condition

We define an approximate closure condition:

$$||i_i - p_i|| < \epsilon$$

for some small threshold ϵ .

Intuitively, this corresponds to trajectories in which the interpretation semantically returns near the original problem region.

Such trajectories may correspond to:

- confirmatory studies,
- replication experiments,

- local theoretical validation,
- or highly incremental scientific progress.

In contrast:

$$||i_i - p_i|| \gg 0$$

indicates trajectories whose interpretations diverge substantially from the original problem framing.

These may correspond to:

- conceptual expansion,
- methodological transfer,
- paradigm generalization,
- or broader epistemic reframing.

13.3 Novelty Metric

We define the empirical novelty score:

$$\mathcal{N}(i) = ||i_i - p_i||$$

which measures the semantic displacement between the initial problem framing and the final interpretation framing of a scientific trajectory.

Importantly, this metric does not measure truth, correctness, or scientific quality. Instead, it approximates the degree of semantic and conceptual displacement induced by the discourse trajectory.

13.4 Future Problem Approximation

A more complete approximation of epistemic closure may be constructed using conditional transition distributions.

Let:

$$\mathcal{D}(P \mid i)$$

be the empirical distribution of problem embeddings associated with semantic neighborhoods around interpretation state i .

We define the expected future problem embedding:

$$\bar{p}_i = \mathbb{E}[P \mid i]$$

which may be estimated operationally using nearest-neighbor averaging over problem embeddings associated with neighboring interpretation states.

This yields an approximate net epistemic displacement:

$$\Delta_E^{(i)} = \bar{p}_i - p_i$$

where:

$$||\Delta_E^{(i)}||$$

represents an estimate of the future problem displacement induced by the discourse trajectory.

14 Theoretical Interpretation Layer

The empirical framework described above constructs semantic representations from scientific discourse without requiring that embedding geometry correspond directly to literal epistemic structure. However, the observed regularities motivate a higher-level theoretical interpretation in which scientific activity is represented as a temporally evolving higher-order epistemic structure.

Importantly, EDEL distinguishes between:

- (a) an *empirical ontology*, in which semantic and topological structures are reconstructed from scientific discourse data, and
- (b) a *generative ontology*, in which these structures become operational substrates for synthetic epistemic agents capable of generating and composing scientific reasoning trajectories.

14.1 EDEL as a Layered Formal Structure

We define empirical EDEL as the pair:

$$\hat{\mathcal{E}} = (\hat{\mathcal{E}}, \hat{\mathcal{T}})$$

where:

- $\hat{\mathcal{E}}$ defines the semantic embedding substrate,
- $\hat{\mathcal{T}}$ defines the temporally evolving epistemic-topological structure reconstructed from discourse.

The semantic embedding structure is defined as:

$$\hat{\mathcal{E}} = (E, \phi)$$

where:

- $E \subseteq \mathbb{R}^d$ is the semantic embedding space,
- $\phi : \mathcal{T}_{disc} \rightarrow E$ maps discourse-role segments into semantic coordinates.

Given a scientific artifact a_i , segmentation produces discourse-role embeddings:

$$p_i, m_i, f_i, i_i \in E$$

corresponding respectively to problem, method, finding, and interpretation discourse.

Importantly, these embeddings are treated as empirical semantic representations rather than direct epistemic states.

14.2 Discourse Simplices

Each scientific artifact induces a discourse simplex:

$$\sigma_i = (p_i, m_i, f_i, i_i)$$

interpreted geometrically as a 3-simplex embedded in semantic space.

Within this interpretation:

- vertices correspond to discourse-role embeddings,

- edges correspond to pairwise discourse transitions,
- triangular faces correspond to partial epistemic configurations,
- the tetrahedral simplex represents a complete discourse trajectory associated with a research approach.

Unlike conventional graph-based representations, discourse simplices preserve higher-order relational structure between all discourse components simultaneously.

14.3 Epistemic Simplicial Complex

The global epistemic organization is represented as a temporally indexed simplicial complex:

$$\mathcal{T} = \{\mathcal{K}_t\}_{t \in \mathbb{T}}$$

where each:

$$\mathcal{K}_t = (\mathcal{S}_t, \mathcal{D}_t)$$

contains:

- a set of discourse simplices:

$$\mathcal{S}_t = \{\sigma_i\}$$

- a family of conditional transition operators:

$$\mathcal{D}_t(Y \mid x), x \in X$$

defined over discourse spaces:

$$X, Y \in \{P, M, F, I\}$$

The transition operators define asymmetric semantic attachment structure between discourse regions.

For example:

$$\mathcal{D}(M \mid p)$$

describes methods empirically associated with neighboring problem states, while:

$$\mathcal{D}(P \mid m)$$

describes problem regions empirically associated with neighboring methods.

Importantly:

$$\mathcal{D}(M \mid p) \neq \mathcal{D}(P \mid m)$$

in general.

The resulting structure therefore defines a directed higher-order epistemic topology rather than a purely symmetric simplicial complex.

14.4 Empirical Interpretation

Within the empirical ontology, the simplicial complex is interpreted as a statistical reconstruction of scientific discourse organization.

In this setting:

- semantic geometry approximates discourse similarity,
- simplicial attachment approximates higher-order epistemic organization,
- transition operators approximate empirical semantic affordances between discourse regions.

Importantly, the framework does not claim that embeddings or simplices constitute literal epistemic states. Rather, they provide computable approximations of structural regularities observable in scientific discourse corpora.

14.5 Closure and Epistemic Expansion

The displacement:

$$\Delta_E(i) = i_i - p_i$$

defines the net semantic displacement between the initial problem discourse and the final interpretive discourse of a scientific artifact.

Its norm:

$$\mathcal{N}(i) = \|i_i - p_i\|$$

is interpreted as an empirical approximation of epistemic expansion.

Small values correspond to semantically closed trajectories in which interpretive discourse remains close to the original problem framing, while large values correspond to semantically expansive trajectories generating displaced conceptual regions.

Higher-order geometric observables may also be defined over discourse simplices, including:

- edge lengths,
- face areas,
- simplex volumes,
- operator-space simplices constructed from displacement operators.

These quantities characterize increasingly complex forms of epistemic transformation geometry.

15 Generative Interpretation Layer

We define generative EDEL as:

$$\mathfrak{E} = (\mathcal{E}, \mathcal{T}, \mathcal{C})$$

where:

$$\mathcal{C} = (\mathcal{O}, \mathcal{M}, \circ)$$

defines a compositional epistemic structure.

Here:

- $\mathcal{O} = \{P, M, F, I\}$ defines discourse-role objects,
- $\mathcal{M}$ defines epistemic morphisms between discourse spaces,
- $\circ$ defines compositional structure over epistemic transformations.

In this interpretation:

$$P \rightarrow M \rightarrow F \rightarrow I$$

is treated as a compositional scientific reasoning trajectory rather than merely an observed semantic sequence.

The conditional transition operators become generative affordance structures.

For example:

$$m^* \sim \mathcal{D}(M \mid p)$$

defines admissible methodological continuations for a problem state p .

Similarly:

$$f^* \sim \mathcal{D}(F \mid m^*)$$

and:

$$i^* \sim \mathcal{D}(I \mid f^*)$$

define generative transitions through epistemic discourse spaces.

Agents may therefore construct novel discourse simplices not explicitly observed in the empirical corpus:

$$\sigma^* = (p, m^*, f^*, i^*)$$

which can subsequently attach to the evolving epistemic complex.

Within this ontology, scientific discovery is interpreted as growth and restructuring of the epistemic simplicial complex through compositional epistemic operations.

Importantly, the category-theoretic layer is not directly observable empirically. Rather, it provides a formal operational semantics for generative scientific systems built on top of empirically reconstructed epistemic topology.

15.1 Epistemic Value

The original Epistemic Landscape (EL) framework models scientific exploration as search over a landscape whose topology represents research approaches and whose height corresponds to their epistemic value. In EDEL, this idea is generalized from isolated points to higher-order epistemic structures represented as discourse simplices.

Formally, let:

$$\sigma_i = (p_i, m_i, f_i, i_i)$$

denote the discourse simplex associated with scientific artifact i , where:

- p_i is the problem discourse embedding,
- m_i is the method discourse embedding,
- f_i is the finding discourse embedding,

- i_i is the interpretation discourse embedding.

We define the epistemic value operator:

$$\mathcal{V} : \mathcal{S} \rightarrow \mathbb{R}$$

where:

$$\mathcal{S} = \{\sigma_i\}$$

is the set of discourse simplices composing the EDEL complex.

The quantity:

$$\mathcal{V}(\sigma_i)$$

represents the epistemic utility associated with the corresponding scientific reasoning configuration. Importantly, epistemic value is treated as formally distinct from semantic geometry and topological structure. A simplex may occupy a dense or central region of the epistemic complex while still having low epistemic utility, and conversely highly valuable simplices may emerge in sparse or weakly connected regions.

This distinction separates:

- the *structure* of scientific discourse,
- from the *value* of exploring particular epistemic regions.

Empirical EDEL In empirical EDEL, epistemic value cannot be observed directly and must therefore be approximated through external scientometric or social indicators. Let:

$$\hat{\mathcal{V}}(\sigma_i)$$

denote an empirical estimator of epistemic value.

Examples include:

- citation counts,
- field-normalized citations,
- PageRank-style influence metrics,
- theorem reuse frequency,
- downstream clinical or technological adoption.

In the present work, we primarily consider:

$$\hat{\mathcal{V}}(\sigma_i) = \log(1 + c_i)$$

where c_i denotes the citation count associated with artifact i .

These quantities should not be interpreted as direct measures of truth or intrinsic scientific merit. Instead, they provide imperfect observational proxies for the social and epistemic impact of scientific artifacts. Epistemic value in empirical EDEL therefore remains partially latent and institutionally mediated.

Generative EDEL In generative EDEL, epistemic value is not inferred from historical traces but defined explicitly through the ground-truth structure of the target problem domain.

Let:

$$\mathcal{G}$$

denote the ground-truth environment associated with a scientific domain or task. Epistemic value is then defined as:

$$\mathcal{V} : \mathcal{S} \times \mathcal{G} \rightarrow \mathbb{R}$$

assigning utility to discourse simplices according to their contribution toward solving or explaining the target environment.

Examples include:

- predictive accuracy,
- theorem proving success,
- simulation performance,
- optimization objectives,
- explanatory compression,
- environment reward.

In this setting, epistemic value becomes operationally well-defined and externally grounded. Scientific agents operating over the EDEL structure can therefore optimize exploration policies relative to explicit utility functions while remaining constrained by the transition structure of the epistemic complex.

Extended EDEL Formalism With the inclusion of epistemic value, the empirical EDEL framework can be summarized as:

$$\hat{\mathfrak{E}} = (\hat{\mathcal{E}}, \hat{\mathcal{T}}, \hat{\mathcal{V}})$$

where:

- $\hat{\mathcal{E}}$ defines the semantic and topological epistemic structure,
- $\hat{\mathcal{T}}$ defines the epistemic transition dynamics,
- $\hat{\mathcal{V}}$ defines the epistemic valuation structure.

Including the compositional semantics layer on the generative ontology yields the extended formulation:

$$\mathfrak{E} = (\mathcal{E}, \mathcal{T}, \mathcal{C}, \mathcal{V})$$

where:

$$\mathcal{C}$$

defines the compositional and categorical structure governing admissible epistemic operations and scientific reasoning processes.